

0.1 RQ2-Specific Coding Codebook

Coding purpose. RQ2 examines how visibly the included reviews treat the components of the Markov decision process used to formulate building-control problems, and how far they identify inconsistent formulation as an obstacle to cross-study comparison.

What the scale measures. The shared 0–3 scale records treatment intensity, meaning how prominently a review treats a coded component or barrier within its own account of the literature. It does not measure whether the formulations used in the primary studies are correct, complete, or well specified, and it does not measure how comparable those studies actually are. A high score indicates that a review makes a component or barrier visible and organizes part of its synthesis around it. The score describes the review, not the field.

Table 1: Shared treatment-intensity scale as applied in RQ2.

Score	Label	General interpretation
0	Not addressed	The component or barrier is not identifiable as a formulation dimension in the review.
1	Mentioned	The component or barrier is named or acknowledged in passing, without substantive development.
2	Discussed	The component or barrier is explained, illustrated, or used in the review’s discussion, but is not systematically classified across the reviewed studies.
3	Structuring role	The component or barrier organizes part of the review’s synthesis, through a taxonomy, an explicit comparative criterion, a cross-study classification, or a recurring interpretive theme.

Structure of RQ2 coding. RQ2 is coded on two separate dimensions, each with its own unit set, plus one survey-level score.

Formulation-component visibility records how visibly a review treats each MDP component across the studies it reviews. Its units are state, action, reward, objectives, and constraints.

Comparability-barrier visibility records how far a review identifies inconsistency in each element as an obstacle to cross-study comparison. Its units are state, action, reward, objective or metric, constraint, baseline, and simulator or testbed. The last two have no counterpart in the first dimension because they concern how studies are evaluated rather than how they are formulated.

The two dimensions are independent. A review can make a component visible without treating its inconsistency as a comparability barrier, and can identify a barrier for an element it does not otherwise classify.

Comparability-problem treatment is a single survey-level score recording how centrally a review treats comparability breakdown as a theme. It is coded directly from the review and is not derived from the barrier unit scores. A review can name several specific barriers while treating comparability as a secondary concern.

Unit vocabulary. The same MDP components are named consistently across both dimensions. Where the survey-level audit tables use the plural forms objectives and constraints and the barrier tables use the singular forms objective or metric and constraint, the labels refer to the same components. Normalization changes labels only and leaves all scores unchanged.

Zero and applicability. A score of 0 means the review does not address the component or barrier. The scale does not separate a component that a review considered and did not develop from one that fell outside its declared scope. Both are recorded as 0. This keeps denominators uniform and is consistent with the construct, which records the treatment a review gives a topic

rather than the reasons for that treatment. A score of 0 should therefore not be read as evidence that a review overlooked something it could have covered.

All coded reviews are scored on both dimensions and on the survey-level score. RQ2 has no applicability route, so the denominator is the full coded corpus.

Table 2: Operational criteria for formulation-component visibility.

Score	Criterion
0	The component is not treated as a visible formulation dimension anywhere in the review.
1	The component is named in passing, for example in a definition of the MDP tuple, without further use in the review’s discussion.
2	The component is explained and used in the review’s discussion of the reviewed literature, but the review does not systematically classify how studies specify it.
3	The component is systematically classified across reviewed studies, used as an explicit comparative criterion, or organized into a taxonomy, table, or figure that structures part of the synthesis.

Operational criteria: formulation-component visibility.

Table 3: Operational criteria for comparability-barrier visibility.

Score	Criterion
0	The review does not identify inconsistency in this element as an obstacle to comparing studies.
1	Inconsistency in this element is noted, but its consequence for cross-study comparison is not developed.
2	The review explains how inconsistency in this element weakens comparison, without organizing its synthesis around that point.
3	Inconsistency in this element is a structuring concern, developed as a recurring theme, a dedicated discussion, or an explicit basis for the review’s conclusions about comparability.

Operational criteria: comparability-barrier visibility.

Table 4: Operational criteria for the survey-level comparability-problem score.

Score	Criterion
0	Comparability across studies is not raised as a problem.
1	Comparability is mentioned as a difficulty without elaboration.
2	The review explains how comparability is weakened, and may contribute partial remedies such as common metrics, but comparability breakdown is not a central structure of the review.
3	Comparability breakdown is a central theme, emphasized across the review or used as a principal basis for its conclusions and recommendations.

Operational criteria: comparability-problem treatment.

Evidence rules. Tables, figures, taxonomies, and dedicated sections are coding evidence, and organizing use of a component or barrier in any of them supports a structuring score. A definition of the MDP tuple in background material is treated as formulation context rather than classification, and on its own supports a score of 2 at most. Repeated mentions do not accumulate into a higher score. Content appearing in the abstract or keywords alone is scored at most 1. Supplementary material was not coded, because its availability is inconsistent across the corpus

and coding it would make reviews non-comparable. Coding reflects the review authors' own account: a formulation choice present in a cited primary study is insufficient unless the review discusses, tabulates, or classifies it. Where evidence falls between two adjacent levels, the lower score is recorded unless the higher criterion is clearly satisfied.

Worked boundary examples. The examples below are taken from the survey-level audit tables that follow.

Summary fields. Each survey-level audit table also carries three narrative fields: formulation strength, the largest inconsistency identified, and the main comparability consequence. These summarize the coded scores in prose and are not themselves coded values. Formulation strength takes a base value of strong or moderate, optionally qualified to indicate the components driving it.

Soft and hard constraints. Constraints are coded as a single unit on both dimensions. The coding does not separate soft constraints implemented as reward penalties from hard feasibility or safety conditions enforced within the formulation, because that distinction was not available as a separate score in the coding as performed. The distinction is nevertheless reported: the RQ2 text separates the two things the constraint scores conflate, noting that most reviews treat constraints through reward penalties, comfort ranges, or storage bounds, while explicit feasibility and safety conditions are treated as a structuring comparison dimension by only one review. The discussion states that the two do not offer equivalent guarantees although the survey literature largely reports them under a single heading. Any future coding pass that scores the two separately would extend this unit rather than replace it, and would require re-reading every review.

Table 5: Worked boundary examples for RQ2 coding.

Review	Unit and dimension	Score	Basis for the boundary
Han et al. (2021)	Constraints, formulation	0	Constraints are not treated as a visible formulation dimension anywhere in the review. Boundary against 1.
Shaqour and Hagishima (2022)	State, formulation	2	States, actions, and rewards are explained in paragraph-level MDP background, which lifts the score above 1, but they do not structure the cross-study review as standalone taxonomies. Boundary against both 1 and 3.
Sierla et al. (2022)	Action, formulation	3	The review sets out to identify action formulations in the literature and classifies studies by binary, discrete, and continuous action types.
Sierla et al. (2022)	Reward, formulation	2	Reward is discussed as the feedback capturing control objectives, but it is not the review’s taxonomy. Contrast with action in the same review.
Pinthurat et al. (2024)	Constraints, formulation	3	Storage constraints are explicit comparative dimensions, including state of charge, state of health, degradation, and operating mode.
Weinberg et al. (2023)	Constraints, formulation	2	Practical limitations such as data frequency, unsafe exploration, and interoperability are discussed, but constraints are not classified across studies. Boundary against 3.
Al-Ani and Das (2022)	Comparability problems, survey level	2	The review contributes common